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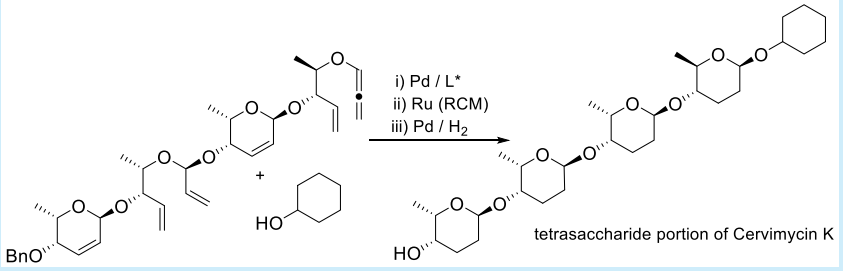
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Convergent Synthesis of Tetrasaccharide Fragment of Cervimycin K

Jihun Kang and Young Ho Rhee*

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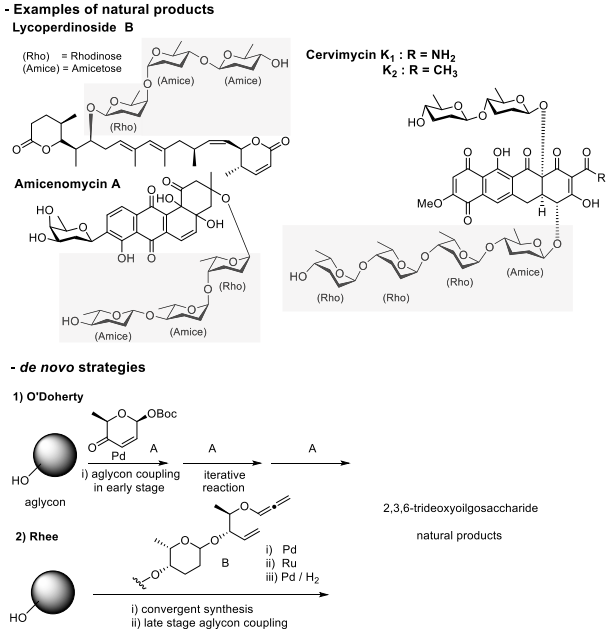
ABSTRACT: A new synthetic approach toward oligosaccharides consisting only of 2,3,6-trideoxypyranoglycosides is reported. The key feature is highlighted by the convergent approach that allows the introduction of the aglycon moiety in the late stage of the synthesis. As an illustrative example, the tetrasaccharide portion of cervimycin K was prepared as cyclohexyl glycoside.

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2-Deoxyoligosaccharides are present in numerous bioactive natural products as a conjugate form to complex aglycon moieties. The length and structure of these oligosaccharides are reported to have a significant effect on the bioactivities of the natural products.[1] Thus, the stereocontrolled synthesis of 2-deoxyoligosaccharides represents one of the most fundamental topics in glycochemistry.[2,3] This topic has been considered very challenging because of the lack of stereodirecting appendages at the 2 position. This problem is particularly well-addressed by oligosaccharides consisting only of 2,3,6-trideoxypyranoglycosides (such as amicetose or rhodinose) linked by both α- or β- anomeric configurations. They constitute the glycosidic part of several natural products such as lycoperdinoside B,[4] amicenomycin A,[5] and cervimycin K[6] (Scheme 1). Despite the relatively short length and simple structure of these oligosaccharides, their stereocontrolled syntheses have been very limited in conventional glycochemistry, because of the poor stability of the anomeric configuration.[7]

Recently, O'Doherty reported a potential solution to this problem based upon the iterative use of the Pd-catalyzed reaction of a pyranose unit (A in Scheme 1).[8−11] Later, Tang further elaborated on the stereodivergent coupling by introducing related catalytic asymmetric transformations.[9] This *de novo* approach represents a significant advance in the synthesis of 2,3,6-trideoxyoligosaccharides. However, incorporation of the method into the total synthesis needs to overcome another hurdle in that structurally complex aglycons should be introduced in an early stage and carried through the subsequent stepwise oligosaccharide synthesis. In this regard, a convergent method that couples the aglycon and glycon portions in a late stage would be highly advantageous. Recently, we reported Pd-catalyzed asymmetric hydroalkox-

Scheme 1. Natural Products Containing 2,3,6-Trideoxyoligosaccharides and Related Synthetic Strategies



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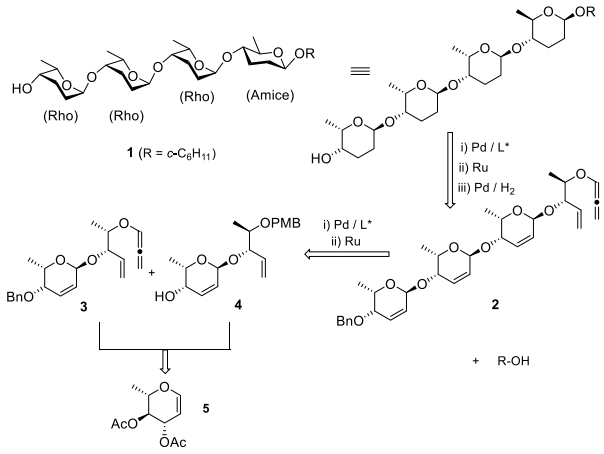
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ylation of ene-alkoxyallenes possessing glycosidic portion (B in Scheme 1).[12,13] This method was successfully evolved into the convergent total synthesis of 11-deoxylandomycin natural products, upon combination with sequential catalysis composed of ring-closing-metathesis (RCM) and Pd-catalyzed hydrogenation. Upon the basis of these precedents, we envisioned that an analogous approach can successfully fulfill the above-mentioned goal in the synthesis of 2,3,6-trideoxyoligosaccharides. As a proof-of-concept example, we report here a convergent synthesis of the tetrasaccharide moiety of antibacterial cervimycin K as a cyclohexyl glycoside.[14,15]

From a retrosynthetic viewpoint, the synthesis of target **1** necessitates preparation of ene-alkoxyallene **2** (Scheme 2). We

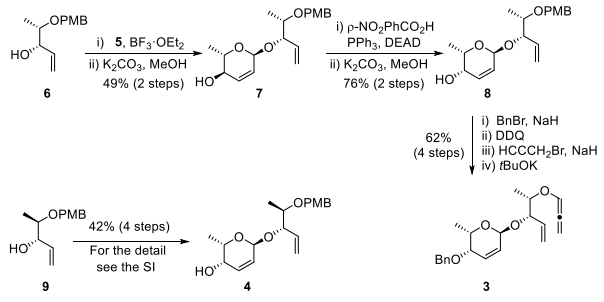
Scheme 2. Retrosynthetic Analysis of Target Structure 1



reasoned that this intermediate can be readily prepared by convergent coupling of **3** with alcohol **4**. Both of these compounds would be then derived from commercially available diacetyl-6-deoxy-(*L*)-glucal **5**.

The synthesis commenced with the preparation of allene **3** (Scheme 3). Reaction of **5** and alcohol **6** in the presence of

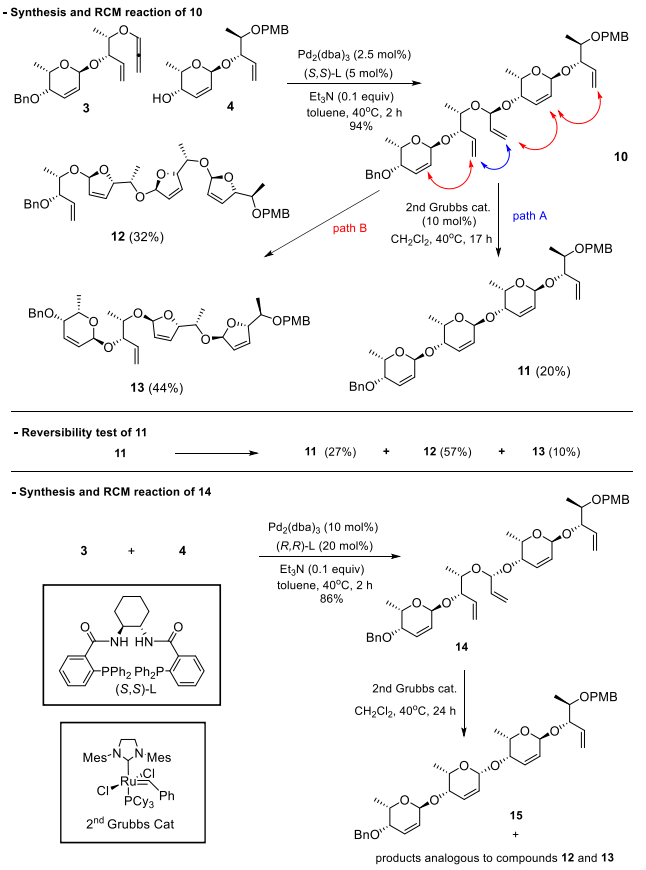
Scheme 3. Synthesis of Allene 3 and Alcohol 4



stoichiometric $BF_3{\cdot}OEt_2$ followed by the deacetylation generated allylic alcohol **7** in 49% yield (over two steps) as a single diastereomer. Inversion of this alcohol under Mitsunobu conditions provided **8** in 76% yield. From this compound, allene **3** was prepared in 62% yield via a four-step event combining (i) benzylation of the alcohol, (ii) chemoselective removal of the PMB group,[16] (iii) propargylation of the resulting alcohol, and (iv) base-catalyzed isomerization. Alcohol **4** was uneventfully prepared from **9** in 42% overall yield using an analogous procedure to that for **8** (for the detailed information, see the SI).

Having prepared **3** and **4**, we then examined the Pd-catalyzed coupling reaction. As depicted in Scheme 4, an initial

Scheme 4. Attempted Synthesis of 11



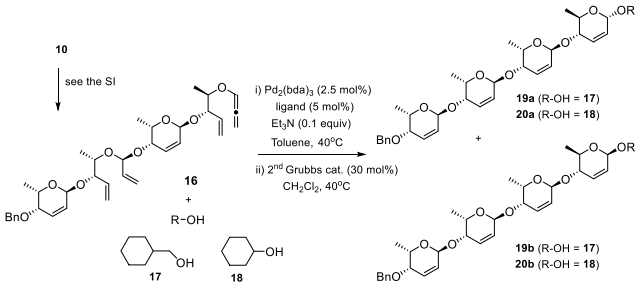
reaction in the presence of $Pd_2(dba)_3$ (2.5 mol %), standard (*S*,*S*)-Trost ligand **L** (5 mol %), and triethylamine (0.1 equiv) in toluene at 40 °C for 2 h generated the acyclic acetal **10** in 94% yield. Even though the diastereoselectivity of this reaction could not be verified at this stage, we decided to see if the RCM reaction of this compound would deliver the desired trisaccharide **11** (path A, Scheme 4). A preliminary reaction using a first generation Grubbs catalyst was very slow. Switching to a second generation catalyst (10 mol %) significantly improved the conversion (>90%). However, the target compound **11** was obtained in only 20% yield. Interestingly, two additional compounds **12** and **13** containing a dihydrofuran skeleton were also obtained in significant 76% combined yield. Apparently, these compounds are formed by the ring-rearrangement metathesis (RRM) of **10** (path B, Scheme 4).[17] Subjection of **11** to similar conditions again produced **12** and **13** as the major products. This result strongly suggests that the products are formed in a reversible manner. To assess the diastereoselectivity of the Pd-catalyzed reaction, we attempted the synthesis of **14** (diastereomeric to **10**) using an enantiomeric (*R*,*R*)-ligand. As depicted in Scheme 4, this reaction turned out to be significantly slower. Complete assumption of the starting materials required the use of 10 mol % $Pd_2(dba)_3$ precatalyst. In this case, **14** was obtained in 86% yield. Because the ^1H NMR spectrum of this compound was indistinguishable from **10**, we performed the RCM reaction of **14**. Again, this reaction produced a mixture of desired **15** and dihydrofuran derivatives analogous to **12** and **13**. Even though

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chromatographic separation of **15** was troublesome, it could be purified by recrystallization from the crude mixture. The ^{1}H NMR spectra of **11** and **15** with respect to the newly generated stereocenter were clearly distinguishable, thus confirming the excellent ligand-driven stereoselectivity in the Pd-catalyzed reactions (for the detailed analysis, see the Supporting Information).

Because the synthesis of trisaccharide moiety **11** by way of RCM reaction was unsuccessful, we decided to carry the acyclic acetal **10** forward without RCM reaction. For this purpose, compound **10** was first converted into allene **16** using the procedure optimized in Scheme 3 (for detailed information, see the Supporting Information). Then, we examined the sequential Pd−Ru catalysis of this compound using primary alcohol **17** or secondary alcohol **18** as the coupling partner. A preliminary reaction of **16** with **17** in the presence of an (*R*,*R*)-ligand under optimized conditions smoothly proceeded to produce the adduct in 93% yield with 21:1 ratio. The subsequent RCM reaction was sluggish. Complete conversion required the use of 30 mol % of a second generation Grubbs catalyst. Nevertheless, the desired tetrasaccharide **19a** was obtained in 92% yield apparently with no competition from the RRM reaction (Table 1).[18] Using an

Table 1. Synthesis of Tetrasaccharides



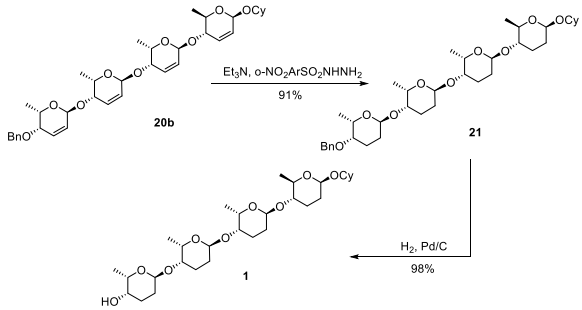
entry	alcohol	ligand	product	yield (%)[a] overall (1st, 2nd)	ratio (*a*:*b*)
1	**17**	(*R*,*R*)-L	**19a**	86 (93, 92)	21:1[b]
2	**17**	(*S*,*S*)-L	**19b**	81 (89, 91)	1:17[b]
3[c]	**18**	(*R*,*R*)-L	**20a**	77 (90, 86)	15:1[d]
4[c]	**18**	(*S*,*S*)-L	**20b**	54 (60, 90)	1:13[d]

[a]Isolated yield. [b]Diastereomeric ratio was determined by crude NMR after hydroalkoxylation. [c]10 mol % $Pd_2(dba)_3$ and 20 mol % of ligand was used. [d]Diastereomeric ratio was determined after RCM reaction.

enantiomeric (*S*,*S*)-ligand reversed the stereoselectivity, generating **19b** possessing all the stereochemical features of the glycosidic part in cervimycin K in 81% yield with high 17:1 selectivity over two steps. In the case of secondary alcohol **18**, the Pd-catalyzed coupling reaction turned out to be slower, presumably due to the increased steric congestion of the substrate. Using a higher loading of $Pd_2(dba)_3$ (10 mol %) was necessary to obtain the acyclic adduct in high 90% yield. The subsequent RCM reaction smoothly proceeded to give **20a** in 86% yield. Finally, diastereomeric **20b**, a key intermediate in our synthesis, was produced in 54% yield (over two steps) with 13:1 selectivity again with the use of an enantiomeric (*S*,*S*)-ligand.

Our final task was to synthesize the target cyclohexyl glycoside **1** from **20b**, which required the execution of two events: (i) hydrogenation of the olefin and (ii) removal of the benzyl group. Initial efforts toward one-pot operation using various Pd catalysts were unsuccessful, because of the extensive decomposition of **20b** possessing labile anomeric centers. Thus, the olefin was first hydrogenated using o-nitrobenzenesulfonyl-hydrazide (NBSH). As depicted in Scheme 5, this method afforded **21** in 91% yield.[19] Removal of the

Scheme 5. Synthesis of Target Compound 1



benzyl group proceeded smoothly by using a Pd/C catalyst under a hydrogen atmosphere, which delivered target compound **1** in 98% yield. The structure of **1** depicted in Scheme 5 was rigorously verified by various two-dimensional NMR spectroscopic techniques. (for detailed information, see the Supporting Information). Overall, tetrasaccharide **1** was assembled by the asymmetric sequential metal catalysis (Pd−Ru−Pd) from commercially available starting material **5**. On the basis of a ligand-driven stereodivergent process, various stereoisomers of **1** can also be accessed with comparable efficiency to that for the naturally occurring structure.

In summary, we developed a new synthetic strategy toward 2,3,6-trideoxyoligosaccharide. Remarkably, the convergent nature makes the method particularly amenable to the total synthesis of the related natural products. As an illustrative example, the first synthesis of tetrasaccharide moiety in cervimycin K has been achieved. Currently, we are working on expanding of the scope of our *de novo* strategy to other oligosaccharides as well as on the total synthesis of related natural products. The results of this study will be reported in due course.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c01404.

Experimental data and copies of ^{1}H and ^{13}C spectra for all new compounds (PDF)

FAIR data, including the primary NMR FID files, for compounds **1**, **3**, **4**, **7**, **8**, **10**−**16**, **19**−**21**. (ZIP)

■ AUTHOR INFORMATION

Corresponding Author

Young Ho Rhee − *Department of Chemistry, POSTECH, Pohang, Kyungbuk, Republic of Korea*; orcid.org/0000-0002-2094-4426; Email: yhrhee@postech.ac.kr

Author

Jihun Kang − *Department of Chemistry, POSTECH, Pohang, Kyungbuk, Republic of Korea*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.1c01404

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Notes

The authors declare no competing financial interest.

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